

Dedication

To all the dreamers who work tirelessly to turn their ambitions into reality. This book is for you.

Acknowledgments

I would like to express my heartfelt gratitude to my parents, without whose support, I would not have been able to reach a point where I could accomplish all of this. Their encouragement has been the foundation of everything I've achieved. A special thanks to Arshad Javed Minhas and Amer Ali, who encouraged me to become a PMP and inspired me to write this book.

Preface

Project management has always been a fascinating field for me, and over the years, I've seen how a structured approach can turn the most complex projects into achievable milestones. Writing this book was a way to share my experiences and provide a straightforward guide for PMP candidates.

I've kept the content concise yet comprehensive, ensuring it covers everything you need to prepare for the PMP exam effectively. This book is designed to simplify your learning process, provide clarity on key concepts, and act as a quick reference material during your preparation journey.

My hope is that this book becomes your go-to resource for not just passing the PMP exam but also enhancing your project management skills in real-world scenarios.

Happy learning and best of luck!

Introduction

Welcome to *Mastering PMP: A Complete Roadmap to Success!*

Whether you're a seasoned professional or just starting out in project management, this book is crafted to guide you through every step of preparing for the PMP exam.

What makes this book unique is its focus on simplicity and relevance. Each topic is broken down into manageable summaries, with subtopics clearly outlined to help you prioritize your studies. The content is aligned with the latest PMP exam syllabus, ensuring you're well-prepared for what's ahead.

Here's what you can expect:

- **Comprehensive Coverage:** All PMP topics and subtopics are included to give you a complete understanding.
- **Efficient Preparation:** Clear and concise content that makes studying easier and faster.
- **Versatility:** A valuable resource for both beginners and experienced professionals.

- **Real-World Application:** Even if you're not taking the PMP exam, the concepts in this book will help you improve how you manage tasks and projects in your daily work.

This book is structured to help you make the most of your time and efforts. Use it as a guide, a reference, and a companion on your path to PMP certification—and beyond. Let's get started!

Table of Contents

Chapter 1: Project Fundamentals	7
Chapter 2: Stakeholders.....	12
Chapter 3: Communication Management	15
Chapter 4: Teams	18
Chapter 5: Risk in a Project	23
Chapter 6: Handling the Issue Log	28
Chapter 7: Quality	30
Chapter 8: Procurement	34
Chapter 9: Scope.....	38
Chapter 10: Project Schedule.....	42
Chapter 11: Project Costs	46
Chapter 12: Agile	51
Chapter 13: Project Closure.....	56

Chapter 1: Project Fundamentals

a. Understanding Projects

- ✓ A project is a temporary activity initiated to increase business value, aligning with the organization's strategy, having a definite start and end.

b. Operations

- ✓ Operations are ongoing activities. Once projects are completed, their outcomes are transferred to the operations team for maintenance and continuity.

c. Programs and Portfolios

- Program:** A collection of related projects.
- Portfolio:** A collection of projects, programs, and operations, managed together to achieve strategic objectives.

d. Organizational Structures

i. Functional Organization

- ✓ The functional manager retains **100% control**.
- ✓ A project expeditor has **no authority** in this structure.

ii. Project-Oriented Organization

- ✓ The project manager holds **complete control** over resource management.

iii. Matrix Organizations

- ✓ **Weak Matrix:** Functional manager has more control, but a project coordinator has more authority compared to a project expeditor.
- ✓ **Strong Matrix:** The project manager is in charge and holds significant control.
- ✓ **Balanced Matrix:** Both the project manager and the functional manager share equal control. Conflicts are escalated to the sponsor for resolution.

iv. The Role of PMO

The PMO (Project Management Office) functions in three key roles:

- ✓ **Supportive:** Provides tools, techniques, and training.
- ✓ **Controlling:** Conducts audits alongside support.
- ✓ **Directive:** Guides the project manager's daily activities.

e. Project Documents

i. Pre-Project Documents

- ✓ **Business Case:** Focuses on tangible, monetary aspects.
- ✓ **Benefit Management Plan:** Covers intangible benefits.

ii. Project Charter

- ✓ The project charter is the first document, officially initiating the project.
- ✓ Approved by the sponsor, it grants authority to the project manager to execute the project.

iii. Key Rules for Pre-Project Documents

- ✓ Once approved, the **project charter**, **business case**, and **benefit management plan** cannot be changed. Modifications require starting a new project.

f. Governance and Risk Management

i. Governance

- ✓ The governance board decides whether a project proceeds or stops. If the business case becomes invalid, the project may be cancelled.

ii. Risk and Assumptions

- ✓ Compliance requirements are documented as **negative risks** in the risk register.
- ✓ The **assumption log** contains a list of assumptions and constraints.

g. Project Management Process Groups

Project work is carried out across **five process groups**:

i. Initiation

Create the following documents:

- ✓ **Project Charter**
- ✓ **Assumption Log**
- ✓ **Stakeholder Register**

ii. Planning

- ✓ Plan the entire project by preparing 18 management plans.
- ✓ Determine whether the project will follow a **predictive** or **agile life cycle**.

iii. **Execution**

- ✓ Perform project work.
- ✓ Train team members and resolve conflicts.
- ✓ Manage issues as they arise.

iv. **Monitoring and Controlling**

- ✓ Compare the project plan against actual performance.
- ✓ Raise change requests via **Integrated Change Control** if plans require updates.
- ✓ Obtain change request approval from the Change Control Board.

v. **Closing**

- ✓ Formally close the project after deliverables are accepted (or rejected).

Chapter 2: Stakeholders

a. Identify Stakeholders

- ✓ Use expert judgment, focus groups, workshops, organizational process assets (OPA), and enterprise environmental factors (EEFs) to identify stakeholders.

b. Stakeholder Register

- ✓ Record identified stakeholders in the stakeholder register.

c. Stakeholder Analysis

- ✓ Categorize stakeholders as **upward**, **downward**, or **sideward** during stakeholder analysis.

d. Prioritize Stakeholders

- ✓ Use tools such as the **power-interest matrix**, **stakeholder cube**, or **salience model** to prioritize stakeholders effectively.

e. Stakeholder Register Review

- ✓ Once the stakeholder register is updated and finalized, the initiation phase is considered complete.

f. Stakeholder Engagement Plan

- ✓ Develop a stakeholder engagement plan that includes strategies to effectively engage stakeholders.

g. Stakeholder Engagement Matrix (SEM)

- ✓ Use the SEM to compare each stakeholder's **current engagement state** with their **desired engagement state** as part of the engagement plan.

h. Approval and Implementation

- ✓ Obtain sponsor or Change Control Board (CCB) approval for the stakeholder engagement plan and begin implementing it.

i. Plan Review

- ✓ Regularly review the engagement plan to ensure its effectiveness.
- ✓ If inadequacies are identified, raise a change request to update the plan.

j. Managing New Stakeholders

- ✓ Add newly identified stakeholders to the stakeholder register.
- ✓ Raise a change request to update related plans, such as the stakeholder plan, communication plan, and resource management plan.

k. Update Assumption Log

- ✓ Ensure the assumption log reflects any changes related to stakeholders.

Chapter 3: Communication Management

a. Communication Management Plan

- ✓ All project communications will adhere to the communication management plan.

b. The 4Rs of Communication

- ✓ The plan ensures the **right information** is delivered at the **right time** to the **right people** to achieve the **right impact**.

c. Scope of the Communication Plan

The plan addresses:

- ✓ Who needs the information?
- ✓ When do they need it?
- ✓ Why do they need it?
- ✓ How will it be delivered?
- ✓ How will it be stored?

- ✓ How and when will it be discarded, including specifying the artifacts involved?

d. Stakeholder Communication Needs

- ✓ Identify and document communication needs for each stakeholder, considering cultural sensitivities, political awareness, and specific communication requirements.

e. Face-to-Face Communication

- ✓ Prefer face-to-face communication wherever possible.
- ✓ For virtual teams facing communication challenges, arrange for co-location at least once or for a specific meeting.

f. Communication Methods

- ✓ Use **pull**, **push**, and **interactive** communication methods or a combination of these as required by the situation.

g. Awareness in Communication

- ✓ Consider political awareness and cultural sensitivities during every communication.

h. Work Performance Data and Information

- ✓ **Work performance data:** Raw data generated during execution.
- ✓ **Work performance information:** Processed data derived during monitoring and control.

i. Work Performance Reports

- ✓ The project manager consolidates data and information into **work performance reports** as part of integration.
- ✓ These reports are distributed to stakeholders following the communication management plan.

j. Virtual Teams

- ✓ While virtual teams offer benefits, challenges like miscommunication must be addressed using tools such as **interactive boards** and **information radiators**.

Chapter 4: Teams

a. Identifying Resources and Skills

- ✓ Determine the required resources and skills for the project by conducting a SWOT analysis and reviewing project requirements.

b. Team Type Selection

- ✓ Decide whether to form a **virtual team**, **physical team**, or **hybrid team** by evaluating the pros and cons of each option.

c. Team Recruitment

- ✓ Hire team members based on the skills required for the project.

d. Team Charter Creation

- ✓ Develop a team charter that defines ground rules, conflict resolution procedures, decision-making processes, and the project's vision.

e. Forming Stage

- ✓ In the initial stage, provide the team with clear directions to start their work.

f. Storming Stage

- ✓ Address conflicts within the team, acting as a coach.
- ✓ Use **active listening** and **emotional intelligence** to understand the conflict's root cause.
- ✓ Apply appropriate conflict management techniques based on the five levels of conflict.

g. Norming Stage

- ✓ As the team begins understanding each other's working habits, provide continuous support to maintain productivity and alignment.

h. Motivational Theories

Understand and apply motivational theories such as:

- ✓ **Maslow's Hierarchy of Needs**
- ✓ **Herzberg's Two-Factor Theory**

- ✓ **Theory X and Theory Y**
- ✓ **Halo Effect**
- ✓ **Expectancy Theory**
- ✓ **McClelland's Theory of Needs**

i. Performing Stage

- ✓ At this stage, the team operates efficiently and can work independently with minimal supervision, following a **laissez-faire leadership style**.

j. Team Release

- ✓ Once the project is complete, release the team to take on new projects.

k. Servant Leadership

- ✓ Adopt a servant leader mindset to guide and support the team, acting as a mentor and removing roadblocks throughout the project lifecycle.

l. Training Needs Assessment

- ✓ Continuously identify training needs to support team development.

- ✓ Conduct training sessions for the whole team initially and offer individualized training as the project progresses.

m. Resource Management Plan

Use the resource management plan to:

- ✓ Acquire, develop, manage, and release teams.
- ✓ Budget for and plan training sessions.
- ✓ Recognize and reward project accomplishments.

n. Resource Calendar

- ✓ Define resource availability and scheduling in the resource calendar.

o. RACI Matrix

Use the RACI matrix to clarify:

- ✓ **Responsible:** Who performs the work.
- ✓ **Accountable:** Who makes decisions.
- ✓ **Consulted:** Who provides input.
- ✓ **Informed:** Who is updated on progress.

p. Responsibility Assignment Matrix (RAM)

- ✓ Document who is responsible for each project task.

Chapter 5: Risk in a Project

a. Definition of Risk

- Risk is calculated as $R = P \times I$ (Probability $\times$ Impact).
- For an event to qualify as a risk:
 - ✓ It has not yet occurred.
 - ✓ A plan exists to address it.

b. Developing the Risk Management Plan

- Use **Organizational Process Assets (OPA)**, **Enterprise Environmental Factors (EEF)**, and other relevant data to create a risk management plan.
- Define the following in the plan:
 - ✓ Risk tolerance, thresholds, appetite, attitude, and risk matrix.

c. Approval of the Plan

- ✓ Begin risk identification after the risk management plan is approved.

d. Identifying Risks

- ✓ Use techniques like **expert judgment**, focus groups, workshops, OPA, EEFs, and prompt lists to identify risks.
- ✓ Document identified risks in the **Risk Register**.

e. Risk Analysis

- ✓ **Qualitative Risk Analysis**: Prioritize risks based on probability and impact.
- ✓ **Quantitative Risk Analysis** (if required): Further refine risk prioritization using numerical data.

f. Developing Risk Responses

- Prepare responses for both:
 - ✓ **Negative Risks (Threats)**: Avoidance, mitigation, and transfer.
 - ✓ **Positive Risks (Opportunities)**: Exploitation, enhancement, and sharing.

g. Proactive Risk Responses

- ✓ Take immediate action to address identified risks proactively.

h. Reactive Risk Responses

- ✓ Use strategies like risk acceptance, where risks, their triggers, and contingency plans are identified, and risk owners are assigned.

i. Risk Escalation

- ✓ Escalate risks (positive or negative) that exceed threshold limits to appropriate authorities.

j. Trigger Occurrence and Response

- ✓ When a risk trigger occurs, the risk owner implements the response plan, and the effectiveness of the response is measured.

k. Residual Risk

- ✓ Evaluate and measure the risks that remain after implementing response measures.

l. Secondary Risk

- ✓ Assess risks that arise as a result of implementing responses to other risks.

m. Risk Audit

- ✓ Conduct audits to evaluate the overall effectiveness of the risk management process.

n. Risk Review Meetings

- ✓ Assess remaining reserves and document their status in the **Risk Report**.

o. Unplanned Events

- ✓ Events that occur without a plan are logged in the **Issue Log**.
- ✓ Issues are always **negative events**, while **missed opportunities** are considered positive issues.

p. Documentation in Risk Register

- ✓ Document all identified risks exclusively in the **Risk Register**.
- ✓ Risks do not require a change request for documentation.

q. Distinction Between Risks and Issues

- ✓ Risks (potential future events) are recorded in the **Risk Register**.

- ✓ The **Risk Management Plan** does not contain specific risks or their responses.
- ✓ Negative events without a plan are logged in the **Issue Log**, which is strictly for adverse occurrences.

Chapter 6: Handling the Issue Log

a. Recording the Issue

- ✓ When something unexpected occurs that was not planned, it must be recorded in the **Issue Log**.

b. Root Cause Analysis

- ✓ Conduct a **root cause analysis** to understand why the issue was not identified and addressed during the planning phase.

c. Change Request Evaluation

- ✓ Assess whether a **change request** is necessary to resolve the issue.
- ✓ If required, submit a **change request** through the formal change management process.

d. Implementation of Response

- ✓ Upon approval of the change request, implement the response and update the **Issue Log** with details of actions taken.

e. Risk Register Update

- ✓ If the issue has the potential to reoccur within the same project, add it to the **Risk Register** with appropriate mitigation strategies.

f. Lessons Learned

- ✓ Document the insights gained from managing the issue in the **Lessons Learned Register** for application in future projects.

Chapter 7: Quality

a. Plan Quality Management

Develop the **Quality Management Plan** to document quality standards and specify the **Cost of Quality (CoQ)**.

- ✓ **Cost of Conformance:** Costs for preventing defects (e.g., training, testing).
- ✓ **Cost of Non-Conformance:** Costs incurred due to defects (e.g., rework, warranty claims).

b. Quality Metrics

- ✓ Define **Key Performance Indicators (KPIs)** to measure critical aspects of the project deliverables and processes.

c. Quality Assurance (Execution Phase)

- ✓ Ensure that **processes** are being followed correctly to meet quality standards.
- ✓ Quality assurance focuses on **process adherence** rather than deliverables inspection.

d. Control Quality (Monitoring and Controlling Phase)

- ✓ Inspect deliverables against the agreed-upon quality standards to verify compliance.
- ✓ If standards are met, produce **verified deliverables**.

e. Tools for Quality Management

- **Inspection:** Examine deliverables to ensure they meet requirements.
- **Audit:** Evaluate whether the processes are being followed effectively.

Types:

- ✓ **First-party:** By the project team.
- ✓ **Second-party:** By another internal team.
- ✓ **Third-party:** By an external, independent body.

f. Sampling

Examine a subset of items for quality instead of the entire population.

- ✓ **Random Sampling:** No specific pattern.
- ✓ **Statistical Sampling:** Uses predefined statistical methods.

g. Quality Tools and Charts

- ✓ **Histogram:** Frequency distribution visualization.
- ✓ **Scatter Diagram:** Identifies correlation between two variables.

- ✓ **Fishbone Diagram:** Also known as **Ishikawa Diagram** or **Cause-and-Effect Diagram**, it helps identify root causes of problems.
- ✓ **Control Charts:** Monitor process stability; investigate when data points fall outside control limits.
- ✓ **Pareto Chart:** Uses the 80/20 principle to prioritize key factors affecting quality.
- ✓ **Checklists:** Verify process adherence.
- ✓ **Check Sheets:** Record and count defects for analysis.

h. Continuous Improvement

- **PDCA Cycle (Plan-Do-Check-Act):** A four-step iterative method for improving processes.
- **Six Sigma:** A methodology aimed at reducing defects to **3.4 per million opportunities**, using the **DMAIC** framework:
 - ✓ **Define** the problem.
 - ✓ **Measure** current performance.
 - ✓ **Analyse** root causes.
 - ✓ **Improve** the process.
 - ✓ **Control** and sustain improvements.

i. Philosophy

- ✓ Quality is not a one-time effort; it's a **continuous improvement process** integral to successful project management.

Chapter 8: Procurement

a. Determine Scope Type

i. Fixed Scope:

Suitable for well-defined requirements.

Contract types:

- ✓ **Firm Fixed Price (FFP):** Fixed amount regardless of costs incurred.
- ✓ **Fixed Price Incentive Fee (FPIF):** Seller earns extra incentives for achieving specific targets.
- ✓ **Fixed Price with Economic Price Adjustment (FP-EPA):** Adjusts for inflation or economic changes.

Risk lies with the **seller**.

ii. Variable Scope:

Used for unclear or evolving requirements.

Contract types:

- ✓ **Cost Reimbursement (CR):** Buyer reimburses seller for actual costs.
- ✓ **Cost Plus Incentive Fee (CPIF):** Incentives for meeting performance targets.
- ✓ **Cost Plus Award Fee (CPAF):** Rewards for achieving defined goals.

Risk lies with the **buyer**.

iii. **Time and Material (T&M) Contracts:**

- ✓ Best for ongoing services or subscriptions.
- ✓ Combines fixed hourly/daily rates with materials.
- ✓ Highest risk lies with the **buyer**.

b. Procurement Management Plan

- ✓ Develop a plan detailing how procurement activities will be conducted, monitored, and controlled.

c. Procurement Strategy

- ✓ Define the approach, tactics, and timelines for procurement.

d. Procurement Statement of Work (PSOW)

- ✓ A document detailing what is required, delivery timelines, and quantities. Shared with potential sellers.

e. Bidder Conference

- ✓ A meeting where sellers can clarify requirements and ask questions.
- ✓ Sellers submit proposals afterward.

f. Contract Negotiation and Signing

- ✓ Negotiate terms and finalize the contract.
- ✓ Once signed, the contract becomes legally binding.

g. Audit and Inspection

- ✓ If organizational audits are required, ensure they're included in the contract.
- ✓ Inspect deliverables for quality and compliance before payment and closure.

h. Conflict Resolution in Procurement

Preferred order of resolution:

- ✓ Seek a **win-win** solution.
- ✓ Adhere to the **contract terms**.
- ✓ Engage a **mediator/arbitrator**.
- ✓ If all else fails, resort to **legal action** (court).

i. Payment and Closure

- ✓ Accept deliverables after inspection.
- ✓ Pay sellers and officially close the procurement process.
- ✓ Note that procurements can be closed at various stages, not just project completion.

j. Termination or Breach

- ✓ Party breaching or terminating the contract is liable for fees or penalties.

Chapter 9: Scope

a. Requirement Management Plan

- ✓ Develop a plan outlining how requirements will be identified, documented, managed, and validated throughout the project.

b. Requirement Collection

Use various tools and techniques to gather requirements:

- ✓ Expert Judgement
- ✓ Focus Groups
- ✓ Prototypes
- ✓ Organizational Process Assets (OPA)
- ✓ Enterprise Environmental Factors (EEF)
- ✓ Interviews
- ✓ Value Stream Mapping
- ✓ Context Diagrams

c. Requirement Documentation

Record all requirements in:

- ✓ **Requirement Traceability Matrix (RTM):** Links requirements to project objectives and deliverables.
- ✓ **Requirement Documentation:** Provides detailed descriptions and acceptance criteria.

d. Define Scope

Use the **Scope Management Plan** to differentiate:

- ✓ **Product Scope:** Features and functionalities of the product.
- ✓ **Project Scope:** Work required to deliver the product.

e. Project Scope Statement

- ✓ Develop a formal document describing the project scope, deliverables, constraints, assumptions, and exclusions.

f. Scope Decomposition

Break down the scope into manageable components using **decomposition techniques** to create:

- ✓ **Work Breakdown Structure (WBS):** Hierarchical representation of project work.
- ✓ **WBS Dictionary:** Detailed descriptions of WBS components.

g. Scope Baseline

Includes:

- ✓ **Project Scope Statement**
- ✓ **WBS**
- ✓ **WBS Dictionary**

Requires **formal approval** before being used as the reference point for scope changes.

h. Managing Scope Changes

Any changes to the scope require:

- ✓ **A Change Request.**
- ✓ Follow the **Change Management Plan** to assess, approve/reject, and implement changes.

i. Deliverable Validation

- ✓ Deliverables are considered **validated/accepted** when they meet the scope baseline criteria and receive formal client approval.

Chapter 10: Project Schedule

a. Schedule Management Plan

Develop a plan that includes:

- ✓ The **software** to be used (e.g., Microsoft Project, Primavera).
- ✓ Scheduling methodologies and formats.
- ✓ Procedures for controlling and updating the schedule.
- ✓ Definitions of key milestones and deliverable timelines.

b. Activity Breakdown

Starting from the **Scope Management**, decompose work packages into:

- ✓ **Lists of Activities:** Tasks required to complete the work package.
- ✓ **Activity Attributes:** Details such as duration, dependencies, and resource requirements.
- ✓ **Milestones:** Significant points or events in the schedule.

c. Activity Relationships

Define relationships among activities using dependency types:

- ✓ **Finish-to-Start (FS):** Activity A must finish before Activity B starts.
- ✓ **Start-to-Start (SS):** Activities A and B can start simultaneously.
- ✓ **Start-to-Finish (SF):** Activity B cannot finish until Activity A starts.
- ✓ **Finish-to-Finish (FF):** Both activities must finish together.

d. Activity Dependencies

Identify and verify dependencies:

- ✓ **Mandatory Dependencies:** Inherent to the work (e.g., design must precede development).
- ✓ **Discretionary Dependencies:** Preferred sequences (e.g., based on best practices).
- ✓ **Internal Dependencies:** Within the project team.

- ✓ **External Dependencies:** Outside the project's control (e.g., vendor delivery).

e. Estimate Activity Durations

Use estimation techniques:

- ✓ **Analogous Estimation:** Based on historical data from similar projects.
- ✓ **Parametric Estimation:** Using statistical relationships (e.g., cost per unit).
- ✓ **Bottom-Up Estimation:** Aggregating estimates for smaller components.
- ✓ **Three-Point Estimation:** Considering optimistic, pessimistic, and most likely scenarios.

f. Critical Path Method (CPM)

- ✓ Calculate the **critical path** to identify the sequence of activities that determines the project duration.
- ✓ Develop a **schedule baseline** to serve as a reference for performance tracking.

g. Schedule Compression Techniques

To reduce schedule duration:

- ✓ **Crashing:** Add resources to shorten activities (increases cost).
- ✓ **Fast-Tracking:** Perform activities in parallel (increases rework risk).

h. Resource Optimization Techniques

Balance resource utilization:

- ✓ **Resource Leveling:** Adjust start/finish dates to resolve conflicts (may affect critical path).
- ✓ **Resource Smoothing:** Adjust activities to align with resource availability without affecting the critical path.

i. Change Management

- ✓ **Crashing, fast-tracking, and resource leveling** require a formal **change request**, including updates to the schedule baseline and project plans.

Chapter 11: Project Costs

a. Cost Management Plan

- Develop a plan that includes:
 - ✓ **Currency:** Define the currency to be used for transactions and reporting.
 - ✓ **Tolerance and Threshold Limits:** Specify acceptable cost variances.
 - ✓ **Estimations:**
 - i. **Rough Estimate:** High-level early-stage estimation (e.g., $\pm 50\%$).
 - ii. **Definitive Estimate:** Detailed and precise estimate (e.g., $\pm 10\%$).
 - ✓ **Funding Sources:** Indicate whether the project is self-financed or supported by external funding.

b. Funding Limit Reconciliation

- ✓ Match the project's planned expenditures with funding availability to ensure a balanced flow of funds.

c. Cost Baseline Development

- ✓ **Estimate Costs:** Use techniques like analogous, parametric, bottom-up, or three-point estimation to calculate the cost of each activity.
- ✓ **Add Contingency Reserves:** Costs allocated for identified risks.
- ✓ **Cost Baseline:** Sum of activity costs and contingency reserves. *Cost Baseline = Cost of Activities + Contingency Reserves*

d. Project Budget

- ✓ **Add Management Reserves** to the cost baseline for unidentified risks.

Project Budget = Cost Baseline + Management Reserves

- ✓ Note: Accessing **management reserves** requires a formal change request.

e. Earned Value Analysis (EVA)

- Evaluate project performance by comparing planned and actual progress. Key metrics include:
 - ✓ **Earned Value (EV):** Value of work completed.
 - ✓ **Planned Value (PV):** Budgeted value of planned work.
 - ✓ **Actual Cost (AC):** Actual cost incurred for completed work.

f. Performance Indices

- i. **Schedule Performance Index (SPI):** Measure of schedule efficiency.

$$\text{SPI} = \text{EV}/\text{PV}$$

- ✓ **SPI > 1:** Ahead of schedule.

- ii. **Cost Performance Index (CPI):** Measure of cost efficiency.

$$\text{CPI} = \text{EV}/\text{AC}$$

- ✓ **CPI > 1:** Under budget.

g. Variance Analysis

- i. **Schedule Variance (SV):** Difference between EV and PV.

$$\mathbf{SV=EV-PV}$$

- ✓ **SV = 0:** On schedule.
- ✓ **SV > 0:** Ahead of schedule.
- ✓ **SV < 0:** Behind schedule.

- ii. **Cost Variance (CV):** Difference between EV and AC.

$$\mathbf{CV=EV-AC}$$

- ✓ **CV = 0:** On budget.
- ✓ **CV > 0:** Within budget limits.
- ✓ **CV < 0:** Over budget.

h. Estimate at Completion (EAC)

- i. Use if the project is over budget to forecast the total cost of completion.

$$\mathbf{EAC=BAC/CPI}$$

- ✓ **BAC:** Budget at Completion (original project budget).

ii. **Total Cost Performance Index (TCPI)**

- Determine the cost efficiency needed to complete the project within the remaining budget.

$$\text{TCPI} = \text{BAC} - \text{EV} / \text{BAC} - \text{AC}$$

- ✓ **TCPI > 1:** Indicates the project is in trouble.

Note: **“Scope, schedule and cost baselines together are called performance baselines”**

Chapter 12: Agile

a. Agile Mindset

- ✓ Agile promotes adaptability, collaboration, and incremental progress.
- ✓ Best suited for projects with evolving requirements and phased deliveries.

b. Agile Values (Manifesto)

- ✓ **Customer collaboration** over contract negotiation.
- ✓ **Responding to change** over following a plan.
- ✓ **Working software** over comprehensive documentation.
- ✓ **Individuals and interactions** over processes and tools.

c. Agile Principles

- ✓ Agile's 12 principles expand upon the four core values, emphasizing continuous delivery, stakeholder satisfaction, and team empowerment.

d. 4. Team Dynamics

- ✓ Foster **T-type leadership**: Specialized knowledge with a broad understanding.
- ✓ Encourage empowerment and accountability in team members.
- ✓ Teams self-select tasks and estimate efforts (story points).

e. Agile Estimation and Planning

- ✓ Use techniques like **affinity estimation**, **Fibonacci series**, and **planning poker** to estimate story points collaboratively.
- ✓ Maintain **lightweight estimation** for quick and flexible planning.
- ✓ **Velocity** is unique to each team and should not be compared.

f. Scrum Framework

- ✓ **Scrum Master**: Servant leader removing impediments.
- ✓ **Product Owner**: Represents customer needs, prioritizes the product backlog.

- ✓ **Product Backlog:** Prioritized list of user stories using techniques like **MoSCoW** or **100 Points**.
- ✓ **Sprint:** 1–4 weeks of focused work with a **Sprint Backlog** of selected tasks.
- ✓ **Daily Standups:** 15-minute progress updates focused on yesterday’s work, today’s plan, and impediments.

g. Progress Tracking

- Use **information radiators** in colocated spaces:
 - ✓ **Burn-down charts:** Remaining work.
 - ✓ **Burn-up charts:** Completed work.
 - ✓ **Kanban boards:** Visualize work-in-progress and bottlenecks.

h. Review and Retrospectives

i. Sprint Review:

- ✓ **Demo:** Present completed stories to the customer for acceptance.
- ✓ Unaccepted work returns to the backlog.

- ii. **Sprint Grooming:** Reprioritize backlog based on customer feedback.
- iii. **Sprint Retrospective:** Reflect on sprint performance using steps like setting the stage, gathering data, generating insights, and planning improvements.

i. Large Teams

- ✓ For teams larger than 12 members, use **Scrum of Scrums (SOS)** for better collaboration.

j. Advanced Practices

- i. **Test-Driven Development (TDD):** Write tests before coding, ensuring quality and cleaner code through refactoring.
- ii. **Lean Practices:** Focus on waste reduction, minimizing **lead time** (request to completion) and **cycle time** (start to finish of task). Use **value stream mapping** to identify inefficiencies.
- iii. **Osmotic Communication:** In co-located teams, overheard conversations naturally enhance understanding and collaboration.

iv. **Spike:** Allocate time for handling risks or proofs of concept. Types:

✓ **Architectural Spike:** For exploration during Sprint Zero.

✓ **Risk Spike:** To mitigate potential risks.

Chapter 13: Project Closure

a. Agile Mindset Project Successful / Unsuccessful

- ✓ Determine whether the project has achieved its goals and objectives.

b. Approval from Sponsor

- ✓ Obtain formal approval from the project sponsor to close the project.

c. Measure/Validate Project Benefits

- ✓ Assess the benefits achieved by the project based on the Benefit Management Plan.

d. Procurement Closure

- ✓ Verify that all procurements are closed and attempt to resolve any remaining open items.

e. Final Report

- ✓ Start drafting the final project report summarizing key outcomes and performance.

f. Lessons Learned Register

- ✓ Document lessons learned during the project and update the organizational Lessons Learned Repository.

g. Document Archival

- ✓ Update all project documents and archive them systematically for future reference.

h. OPA Updates

- ✓ Ensure both documents and lessons learned are stored in Organizational Process Assets (OPA).

i. Operations Team Training

- ✓ Confirm the operations team is trained and prepared for the transition of deliverables.

j. Final Report Communication

- ✓ Share the final report with stakeholders according to the Communication Management Plan.

k. Team Release and Administrative Closure

- ✓ Release project team members and complete all administrative closure activities.

l. Celebration and Project Closure

- ✓ Celebrate the project's success and formally close the project.

The End